BALANCE PLATFORM AND INVERTED PENDULUM FOR ROBOTIC VESTIBULAR CONTROL

Bolen, BP[1]; McNeal, JS[1]; and Hunt, AJ[1]
[1]Agile and Adaptive Robotics Lab, Department of Mechanical and Materials Engineering, Portland State University, Portland, OR USA
email: bbolen83@gmail.com

INTRODUCTION

In our lab we are working on biomimetic robots, in part, to gain insight into biological processes. Our robots are actuated with Festo brand Braided Pneumatic Actuators (BPAs) that serve as artificial muscles. When designing to be truly biomimetic, these BPAs are controlled with a Synthetic Nervous System[1], although that is outside the scope of this abstract.

The work presented here is the creation of an inverted pendulum actuated by BPAs that serves an analogous function to a human tibia and foot. This inverted pendulum is to incorporate pseudo-vestibular feedback via an Inertial Measurement Unit (IMU). It had the further requirement of 150 degrees range of motion (RoM). The inverted pendulum was to be placed on a balance platform and maintain balance while the platform oscillated.

METHODS

The balance platform is designed with an 80/20 brand extruded aluminum frame, a 1 DoF aluminum plate for the inverted pendulum to sit on, and motor and power screw to rotate the plate, a potentiometer for plate angle feedback, a 24V power supply, and an Arduino microcontroller to control the movement of the plate (Figure 1).

The inverted pendulum had a base ("foot") and body ("tibia") connected with a single DoF Joint. All constructed with 3D printed Onyx material, save for some extruded aluminum at the bottom of the foot. A 9-axis L3GD20 + LSM303D IMU was attached to the top of the tibia. Proportional pressure control was used to control the BPAs. A novel reverse pulley system was used reach our desired high RoM. The BPAs were mounted to the base and connected the tibia via high strength string acting as artificial tendon (Figure 1).

A second inverted pendulum was created that was more biomimetic than the first version. It had a ball and socket ankle joint actuated with three BPAs, one for plantarflexion and two for dorsiflexion (one inverting and the other everting). The BPAs were mounted to the tibia and controlled with a bang-bang controller sending signals to the valve manifold (Figure 2).

RESULTS AND DISCUSSION

The balance platform was able to oscillate as desired. The inverted pendulum was able to maintain upright balance on the oscillating balance platform. It could achieve the desired RoM.

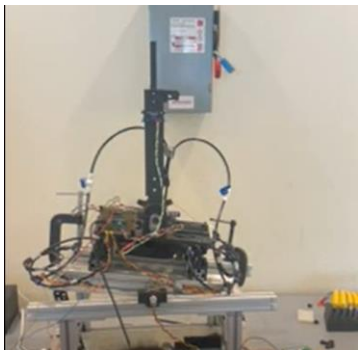


Figure 1: Inverted pendulum balancing on oscillating platform.

The biomimetic inverted pendulum was not tested on the platform, but was able to maintain upright balance with no assistance.



Figure 2: Biomimetic inverted pendulum.

CONCLUSIONS

Further testing can be done with both inverted pendulums and modifying the controllers to use a Synthetic Nervous System.

REFERENCES

1. Hilts WW, et al. *Conference on Biomimetic and Biohybrid Systems*, LCNS **10384**, 2017.

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